EE 533 Information Theory

Week 05

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March 17, 2025

Existence of Canonical Codes (Binary Case)

For any discrete source X there exists an optimal, i.e. with the minimum expected length, instantaneous code $\mathcal{C}$ such that

i) If $p_X(x) > p_X(z)$ then $\ell(\mathcal{C}(x)) \leq \ell(\mathcal{C}(z))$.
ii) The two longest codewords have the same length.
iii) The codewords of the two least likely symbols differ only on their last (least important) digit.

Proof.

i) If not, i.e $\exists x, z$ s.t. $p_X(x) > p_X(z)$, and $\ell(\mathcal{C}(x)) > \ell(\mathcal{C}(z))$, then consider a $\mathcal{C}'$ in which codewords of x and z are swaped:

$$\begin{aligned}\mathbf{E}\left[\ell(\mathcal{C}'(X))\right] &= \mathbf{E}\left[\ell(\mathcal{C}(X))\right] + p_X(x)\left[\ell(\mathcal{C}(z)) - \ell(\mathcal{C}(x))\right] + p_X(z)\left[\ell(\mathcal{C}(x)) - \ell(\mathcal{C}(z))\right] \\ &= \mathbf{E}\left[\ell(\mathcal{C}(X))\right] - [p_X(x) - p_X(z)][\ell(\mathcal{C}(x)) - \ell(\mathcal{C}(z))] \\ &< \mathbf{E}\left[\ell(\mathcal{C}(X))\right]\end{aligned}$$

ii) If not then at least one of them is a leaf that does not have a sibling and thus its codeword can be shortened by one.

iii) If not they can be made sibling with a swap the does not change the expected length.

Optimality of Huffman Codes (Binary Case)

Huffman coding is optimal: if $\mathcal{C}$ is a Huffman code, then for any other uniquely decodable code $\mathcal{C}'$ we have $L(\mathcal{C}) \leq L(\mathcal{C}')$.

Proof.

Without loss of generality assume that X is a r.v. taking values from $\{1, \ldots, m\}$

Let $m+k$ be the symbol obtained by the kth application of Huffman's combining.

Let X_{m-k} be the r.v. obtained by k applications of Huffman's combining.

Let $\mathcal{X}_n$ be the the set of all possible realizations of X_n for $n \in \{1, \ldots, m\}$

e.g.: $\mathcal{X}_{m-1} = \{1, \ldots, (m+1)\} \setminus \{t, s\}$ where t and s are the symbols combined in the firt step.

$\forall n \in \{2, \ldots, m\}$, rvs X_n and X_{n+1} have $n-1$ common realizations: $|\mathcal{X}_n \cap \mathcal{X}_{n+1}| = n-1$

Let a_{n+1} be smaller the element of $\mathcal{X}_{n+1} \setminus \mathcal{X}_n$ and b_{n+1} be the larger element of $\mathcal{X}_{n+1} \setminus \mathcal{X}_n$.

Extension $\mathcal{E}(\mathcal{C}_n)$ of an instantenous code $\mathcal{C}_n$ for X_n is an instantenous code for X_{n+1} s.t.

$$\mathcal{E}(\mathcal{C}_n)(x) = \begin{cases} \mathcal{C}_n(x) & x \in \mathcal{X}_n \cap \mathcal{X}_{n+1} \\ \mathcal{C}_n(2m-n)0 & x = a_{n+1} \\ \mathcal{C}_n(2m-n)1 & x = b_{n+1} \end{cases}$$

Note that $\mathcal{X}_n \setminus \mathcal{X}_{n+1} = \{2m-n\}$

Optimality of Huffman Codes (continued)

$$L(\mathcal{E}(\mathcal{C}_n)) = L(\mathcal{C}_n) + p_{X_{n+1}}(a_{n+1}) + p_{X_{n+1}}(b_{n+1}) \qquad (*)$$

$\forall n \in \{2,\ldots,m\}$, let $\mathcal{C}_n^\star$ be a canonical instantenous code for X_n.

Let ψ_{n-1} be all but the last digit of $\mathcal{C}_n^\star(a_n)$ and $\mathcal{C}_n^\star(b_n)$

Condenstation $\mathcal{K}(\mathcal{C}_n)$ of a canonical instantenous code $\mathcal{C}_n^\star$ is an instantenous code for X_{n-1} s.t.

$$\mathcal{K}(\mathcal{C}_n^\star)(x) = \begin{cases} \mathcal{C}_n^\star(x) & x \in \mathcal{X}_n \cap \mathcal{X}_{n-1} \\ \psi_{n-1} & x \in \mathcal{X}_{n-1} \setminus \mathcal{X}_n \end{cases}$$

$$L(\mathcal{K}(\mathcal{C}_{n+1}^\star)) = L(\mathcal{C}_{n+1}^\star) - p_{X_{n+1}}(a_{n+1}) - p_{X_{n+1}}(b_{n+1}) \qquad (**)$$

For all canonical codes $\mathcal{C}_{n+1}^\star$ and $\mathcal{C}_n^\star$ for $n \in \{2,\ldots,m-1\}$, by (*) and (**) we have

$$\left[L(\mathcal{E}(\mathcal{C}_n^\star)) - L(\mathcal{C}_{n+1}^\star)\right] + \left[L(\mathcal{K}(\mathcal{C}_{n+1}^\star)) - L(\mathcal{C}_n^\star)\right] = 0.$$

$L(\mathcal{E}(\mathcal{C}_n^\star)) \geq L(\mathcal{C}_{n+1}^\star)$ and $L(\mathcal{K}(\mathcal{C}_{n+1}^\star)) \geq L(\mathcal{C}_n^\star)$ imply $L(\mathcal{E}(\mathcal{C}_n^\star)) = L(\mathcal{C}_{n+1}^\star)$ and $L(\mathcal{K}(\mathcal{C}_{n+1}^\star)) = L(\mathcal{C}_n^\star)$

Extension and condensation preserves optimality of X_n's.

Huffman code is the order $m-1$ extension of $\mathcal{C}_1^\star$. Why uniquely decodable? Why not prefix?

For $D \geq 2$: Define the canonical codes using $2+(m-2 \mod D-1)$ longest codewords instead of the two longest codewords and establish the existence the optimal codes that are canonical.

Shannon Codes vs Shannon-Fano-Elias Codes

The existence proof for the Shannon code relies on (extended) Kraft inequality, i.e. Shannon code implicitly orders source symbols according to their probabilities first.

> Without the initial ordering of source symbols according to their probabilities the existence proof fails! **Why? (HW)**

Shannon-Fano-Elias coding works without the ordering but $\ell(x) = \lceil \log \frac{1}{p_X(x)} \rceil + 1$.
Recall the function g mapping finite strings of elements of $\mathcal{D}$ to the half open subintervals of $[0, 1)$ in the existence proof for codes satisfying extended Kraft's inequality

$$g(y_1^n) := \left[0.y_1 \ldots y_n, 0.y_1 \ldots y_n + D^{-n}\right) \qquad \forall y_1^n \in \mathcal{D}^*$$

- $g : \mathcal{D}^* \to \left\{ \left[\frac{s}{D^n}, \frac{s+1}{D^n}\right) : n \in \mathbb{Z}_+, s \in \{0, 1, \ldots, D^n - 1\} \right\}$ is invertible.
- $y_1^n \in \mathcal{D}^*$ is a prefix of $z_1^k \in \mathcal{D}^*$ iff $g(z_1^k) \subset g(y_1^n)$.
- $\mathcal{C} \subset \mathcal{D}^*$ is an instantenous code iff elements of $g(\mathcal{C}) \subset g(\mathcal{D}^*)$ are non-overlapping.

Shannon-Fano-Elias Codes

For each x with a positive $p_X(x)$, half open intervals $[\mathbf{P}(X < x), \mathbf{P}(X \leq x))$ are disjoint but they are not necessarily in $g(\mathcal{D}^*)$.

$$\ell(x) := \lceil \log_D \tfrac{1}{p_X(x)} \rceil + 1 \qquad \Rightarrow p_X(x) \geq D^{1-\ell(x)}$$

$$\mathcal{C}(x) := \left\lfloor \left(\mathbf{P}(X < x) + \tfrac{p_X(x)}{D}\right) D^{\ell(x)} \right\rfloor \qquad \Rightarrow \mathcal{C} \subset \mathcal{D}^*$$

$$\left[\tfrac{\mathcal{C}(x)}{D^{\ell(x)}}, \tfrac{\mathcal{C}(x)+1}{D^{\ell(x)}}\right) \in g(\mathcal{D}^*) \qquad \text{by } \mathcal{C}(x) \in \{0, \ldots, D^{\ell(x)} - 1\}$$

$$\mathcal{C}(x) \geq \lfloor \mathbf{P}(X < x)\, D^{\ell(x)} + 1 \rfloor \qquad \text{by } p_X(x) \geq D^{1-\ell(x)}$$

$$> \mathbf{P}(X < x)\, D^{\ell(x)}$$

$$\mathcal{C}(x) + 1 \leq \left(\mathbf{P}(X < x) + \tfrac{2p_X(x)}{D}\right) D^{\ell(x)} \qquad \text{by } p_X(x) \geq D^{1-\ell(x)}$$

$$\leq \mathbf{P}(X \leq x)\, D^{\ell(x)} \qquad \text{by } D \geq 2$$

$$g(\mathcal{C}(x)) \subset (\mathbf{P}(X < x), \mathbf{P}(X \leq x)) \qquad \Rightarrow \text{ elements of } g(\mathcal{C}) \text{ are non-overlapping.}$$

Then $\mathcal{C} \subset \mathcal{D}^*$ is an instantenous code because $g(\mathcal{C})$ are non-overlapping.

Huffman Code and Other Uniquely Decodable Codes

$\lceil \log \frac{1}{p_X(x)} \rceil$ of the Shannon code can be much larger than the codeword length in the Huffman code.

$$p_X(x) = \begin{cases} 63/64 & x = a \\ 1/64 & x = b \end{cases} \qquad \mathcal{C}_H(x) = \begin{cases} 0 & x = a \\ 1 & x = b \end{cases} \qquad \mathcal{C}_S(x) = \begin{cases} 0 & x = a \\ 100000 & x = b \end{cases}$$

$$L(\mathcal{C}_H) = 1 \qquad L(\mathcal{C}_S) = \frac{69}{64}$$

$\lceil \log \frac{1}{p_X(x)} \rceil$ of the Shannon code can be smaller than the codeword length in the Huffman code.

$$p_X(x) = \begin{cases} 6/16 & x = a \\ 5/16 & x = b \\ 4/16 & x = c \\ 1/16 & x = d \end{cases} \qquad \mathcal{C}_H(x) = \begin{cases} 0 & x = a \\ 10 & x = b \\ 110 & x = c \\ 111 & x = d \end{cases} \qquad \mathcal{C}_S(x) = \begin{cases} 00 & x = a \\ 01 & x = b \\ 10 & x = c \\ 110 & x = d \end{cases}$$

$$L(\mathcal{C}_H) = \frac{31}{16} \qquad L(\mathcal{C}_S) = \frac{33}{16}$$

But on average Huffman code is shorter.

Competitive Optimality of The Shannon Code I

Lemma

If $\ell(x) = \lceil \log_D \frac{1}{p_X(x)} \rceil$ for all source symbols x, then any uniquely decodable code with codeword lengths ℓ' satisfies

$$\mathbf{P}\big(\ell(X) \geq \ell'(X) + c\big) \leq D^{1-c} \qquad \forall c \in \mathbb{Z}_+$$

Proof.

$$\begin{aligned}
\mathbf{P}\big(\ell(X) \geq \ell'(X) + c\big) &\leq \mathbf{P}\Big(1 + \log_D \tfrac{1}{p_X(X)} \geq \ell'(X) + c\Big) \\
&= \mathbf{P}\Big(p_X(X) \leq D^{1-\ell'(X)-c}\Big) \\
&= \sum_{x : p_X(x) \leq D^{1-\ell'(x)-c}} p_X(x) \\
&\leq \sum_x D^{1-\ell'(x)-c} \\
&\leq D^{1-c} && \text{by McMillan inequality}
\end{aligned}$$

Competitive Optimality of The Shannon Code II

Lemma

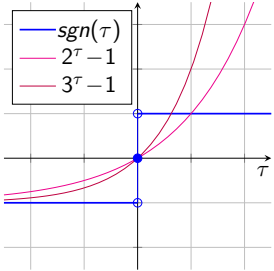
If $\ell(x) = \log_D \frac{1}{p_X(x)}$ for all source symbols x, then any uniquely decodable code with codeword lengths ℓ' satisfies

$$\mathbf{P}\big(\ell(X) < \ell'(X)\big) \geq \mathbf{P}\big(\ell(X) > \ell'(X)\big)\,,$$

with equality iff $\ell(x) = \ell'(x)$ for all x.

Proof.

$$\begin{aligned}
\mathbf{P}\big(\ell(X) > \ell'(X)\big) - \mathbf{P}\big(\ell(X) < \ell'(X)\big) &= \sum\nolimits_x p_X(x)\,\mathrm{sgn}(\ell(x) - \ell'(x)) \\
&\leq \sum\nolimits_x p_X(x)\,(D^{\ell(x)-\ell'(x)} - 1) \\
&= \Big[\sum\nolimits_x D^{-\ell(x)} D^{\ell(x)-\ell'(x)}\Big] - 1 \\
&= \Big[\sum\nolimits_x D^{-\ell'(x)}\Big] - 1 \\
&\leq 0
\end{aligned}$$



Competitive Optimality of The Shannon Code III

Lemma

If $\ell(x) = \lceil \log_D \frac{1}{p_X(x)} \rceil$ for all source symbols x, then any uniquely decodable code with codeword lengths ℓ' satisfies

$$\mathbf{P}\big(\ell(X) < \ell'(X) + 1\big) \geq \mathbf{P}\big(\ell(X) > \ell'(X) + 1\big).$$

Proof.

$$\begin{aligned}
\mathbf{P}\big(\ell(X) > \ell'(X) + 1\big) - \mathbf{P}\big(\ell(X) < \ell'(X) + 1\big) &= \sum\nolimits_x p_X(x)\,\mathrm{sgn}(\ell(x) - \ell'(x) - 1) \\
&\leq \sum\nolimits_x p_X(x)\,(D^{\ell(x)-\ell'(x)-1} - 1) \\
&= \Big[\sum\nolimits_x p_X(x)\,D^{\ell(x)-\ell'(x)-1}\Big] - 1 \\
&\leq \Big[\sum\nolimits_x D^{1-\ell(x)} D^{\ell(x)-\ell'(x)-1}\Big] - 1 \\
&= \Big[\sum\nolimits_x D^{-\ell'(x)}\Big] - 1 \\
&\leq 0
\end{aligned}$$

Generation of Discrete Distributions From Fair Coin Flips

Lossless Source Coding Problem: What is the minimum expected length of binary strings that can represent realizations of a discrete random variable X?

Tools: Binary trees and the binary expansion.

How can we use binary trees to generate discrete random variables from fair coin flips?

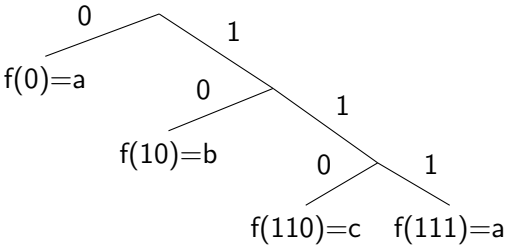
- Take a (possibly infinite) complete binary tree, i.e a binary tree in which each node is either a leaf or has two direct descendants (children).
- $\mathsf{X} = f(\mathsf{Y})$ where $f : \mathcal{Y} \to \mathcal{X}$ and $\mathcal{Y}$ is the set of all of the leaves of the tree.
- Start from the root and at each node choose one of the two direct descendants using a fair coin flip until you reach a leaf $y \in \mathcal{Y}$. Then the probability of reaching any leaf $y \in \mathcal{Y}$ is $2^{-\mathsf{T}(y)}$, where $\mathsf{T}(y)$ is the depth of the leaf, i.e. number of coin flips it takes to get to the leaf from the root.

For any algorithm based on fair coin flips there is a corresponding complete binary tree and a function $f(\cdot)$.

Example

$$p_X(x) = \begin{cases} 5/8 & x = a \\ 1/4 & x = b \\ 1/8 & x = c \end{cases}$$

$$\mathcal{Y} = \{0, 10, 110, 111\}$$



What is the minimum expected number of fair coins needed to generate a discrete random variable X , i.e., What is the minimum expected depth of binary trees that can be used to generate a discrete random variable X?

Lemma

If on a complete binary tree the probability of each leaf $y \in \mathcal{Y}$ is $p_{\mathsf{Y}}(y) = 2^{-\mathsf{T}(y)}$, then the expected the depth of tree is

$$\mathbf{E}\left[\mathsf{T}\right] = H(\mathsf{Y})\,.$$

Proof.

$$\begin{aligned} H(\mathsf{Y}) &= \mathbf{E}\left[\log \frac{1}{p_{\mathsf{Y}}(\mathsf{Y})}\right] \\ &= \mathbf{E}\left[\log \frac{1}{2^{-\mathsf{T}(\mathsf{Y})}}\right] \\ &= \mathbf{E}\left[\mathsf{T}\right]. \end{aligned}$$

□

Lemma

The expected number of fair coin flips necessary to generate a discrete random variable X is greater than or equal to its entropy, that is

$$\mathbf{E}\left[\mathsf{T}\right] \geq H(\mathsf{X})\,.$$

Proof.

Consider the complete tree and the function $f(\cdot)$ corresponding to the algortihm satisfying $\mathsf{X} = f(\mathsf{Y})$ and invoke $H(f(\mathsf{Y})) \leq H(\mathsf{Y})$.

□

Lemma

The expected number of fair coin flips $\mathbf{E}[\mathrm{T}]$ *necessary to generate a a discrete random variable* X *is not larger than* $H(\mathrm{X})+2$,

$$\mathbf{E}[\mathrm{T}] < H(\mathrm{X})+2.$$

Proof.

Obtain a complete tree with leaves y using the binary expansions $p_{\mathrm{X}}(x)=\sum_{t=1}^{\infty} a_t(x)2^{-t}$:

If $a_t(x)=1$ then one leaf of depth t needs to be assigned to x, i.e. $f(y)=x$ for that leaf.

For all $x\in\mathcal{X}$ there exists $n(x)\in\mathbb{Z}_+$ such that $2^{-n(x)}\le p_{\mathrm{X}}(x)<2^{1-n(x)}$.

$$\begin{aligned}
\mathbf{E}[\mathrm{T}]-H(\mathrm{X}) &= \mathbf{E}[\mathrm{T}+\log p_{\mathrm{X}}(\mathrm{X})] && \text{because } H(\mathrm{X})=\mathbf{E}[-\log p_{\mathrm{X}}(\mathrm{X})]\\
&< \mathbf{E}[\mathbf{E}[\mathrm{T}-n(\mathrm{X})|\mathrm{X}]]+1 && \text{by } p_{\mathrm{X}}(\mathrm{X})\le 2^{1-n(\mathrm{X})} \text{ and the law of iterated exp.}\\
\mathbf{E}[\mathrm{T}-n(\mathrm{X})|\mathrm{X}] &= \sum_{t\ge n(\mathrm{X})} \frac{a_t(\mathrm{X})2^{-t}}{p_{\mathrm{X}}(\mathrm{X})}(t-n(\mathrm{X})) && \text{since } \begin{cases} a_t(\mathrm{X})=0\ \forall t<n(\mathrm{X}),\\ p_{\mathrm{X}}(\mathrm{X})=\sum_{t=1}^{\infty} a_t(\mathrm{X})2^{-t}\end{cases}\\
&\le \frac{2^{-n(\mathrm{X})}}{p_{\mathrm{X}}(\mathrm{X})}\sum_{\tau\in\mathbb{Z}_+} 2^{-\tau}\tau && \text{for } \tau=t-n(\mathrm{X})\\
&= \frac{2^{1-n(\mathrm{X})}}{p_{\mathrm{X}}(\mathrm{X})} && \text{by } \sum_{\tau\in\mathbb{Z}_+} 2^{-\tau}\tau=2\\
&\le 1 && \text{by } p_{\mathrm{X}}(\mathrm{X})\le 2^{1-n(\mathrm{X})}
\end{aligned}$$

Lossless Variable Length Source Coding

A lossless variable length source code for X ensures $\mathbf{P}\left(\mathcal{C}^{-1}(\mathcal{C}(\mathsf{X})) = \mathsf{X}\right) = 1$, where

Encoder $\mathcal{C} : \mathcal{X} \to D^*$

Decoder $\mathcal{C}^{-1} : D^* \to \mathcal{X}$

Goal To minimize $\mathbf{E}\left[\ell(\mathcal{C}(\mathsf{X}))\right]$

Motivation For a uniquely decodable $\mathcal{C}$ and independent identically distributed $\mathsf{X}_1, \mathsf{X}_2, \mathsf{X}_3, \ldots$

$$\mathbf{P}\left(\mathcal{C}^{*-1}(\mathcal{C}^*(\mathsf{X}_1^n)) = \mathsf{X}_1^n\right) = 1 \qquad \forall n \in \mathbb{Z}_+$$

$$\frac{\ell(\mathcal{C}^*(\mathsf{X}_1^n))}{n} \xrightarrow{\mathbf{P}} \mathbf{E}\left[\ell(\mathcal{C}(\mathsf{X}))\right] \qquad \text{by W.L.L.N.}$$

If $\mathcal{C}$ is Shannon-Fano-Elias Code on X

$$H(\mathsf{X}) + 2 \geq \mathbf{E}\left[\ell(\mathcal{C}(\mathsf{X}))\right] \geq H(\mathsf{X})$$

If $\mathcal{C}$ is Shannon-Fano-Elias Code on X_1^k

$$H(\mathsf{X}) + \tfrac{2}{k} \geq \tfrac{1}{k}\mathbf{E}\left[\ell(\mathcal{C}(\mathsf{X}_1^k))\right] \geq H(\mathsf{X})$$

Then by the W.L.L.N., for any $k \in \mathbb{Z}_+$ and $\epsilon > 0$

$$\mathbf{P}\left(\left|\tfrac{1}{kn}\ell(\mathcal{C}^*(\mathsf{X}_1^{kn})) - H(X)\right| \leq \tfrac{3}{k}\right) \geq 1 - \epsilon$$

for n large enough.

Fixed Length Source Coding

A length n rate R source code $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$ uses nR bits to send n source symbol:

Encoder $\varphi^{(n)} : \mathcal{X}_1^n \to \{1, 2, \ldots, \lfloor 2^{nR} \rfloor\}$

Decoder $\widetilde{\varphi}^{(n)} : \{1, 2, \ldots, \lfloor 2^{nR} \rfloor\} \to \mathcal{X}_1^n$

Goal To minimize $\mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right)$ where $\hat{\mathrm{X}}_1^n = \widetilde{\varphi}^{(n)}(\varphi^{(n)}(\mathrm{X}_1^n))$

Motivation For i.i.d. $\mathrm{X}_1, \mathrm{X}_2, \mathrm{X}_3, \ldots$ determine the rate R for which there exists $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$'s s.t.

$$\lim_{n\to\infty} \mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right) = 0 \qquad \text{for all } n \text{ large enough.}$$

Lemma (Achievability)

For any $R > H(\mathrm{X})$ and $\epsilon > 0$, $\exists n_0$ s.t. for all $n \geq n_0$ there exist a rate R code s.t. $\mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right) \leq \epsilon$.

Proof.

Use uniquely decodable variable length codes and disregard codewords for the source vectors that require more than R bits per symbol. □

Lemma (Converse)

For any rate R code $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$,

$$\mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right) \geq \left(H(\mathrm{X}) - R - \tfrac{1}{n}\right) \frac{1}{\log|\mathcal{X}|}$$

Converse for Fixed Length Source Coding via Fano's Inequality

Lemma (Converse)

For any $R < H(\mathsf{X})$ and rate R code $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$,

$$\mathbf{P}\left(\hat{\mathsf{X}}_1^n \neq \mathsf{X}_1^n\right) \geq \left(H(\mathsf{X}) - R - \tfrac{1}{n}\right) \frac{1}{\log|\mathcal{X}|}$$

Proof.

Fano's inequality: For any $\mathcal{Y}$ valued discrete random variables Y, Z satisfy

$$H(\mathsf{Y}|\mathsf{Z}) \leq \hbar(P_{\mathbf{e}}) + P_{\mathbf{e}} \log|\mathcal{Y}| \qquad \text{where } P_{\mathbf{e}} = \mathbf{P}(\mathsf{Y} \neq \mathsf{Z}).$$

$$\begin{aligned}
nH(\mathsf{X}) &= H(\mathsf{X}_1^n) && \text{because } \mathsf{X}_t\text{'s are i.i.d.} \\
&= H(\mathsf{X}_1^n, \mathsf{Z}) && \text{for } \mathsf{Z} := \widetilde{\varphi}^{(n)}(\varphi^{(n)}(\mathsf{X}_1^n)) \text{ by chain rule \& } H(g(\mathsf{Y})|\mathsf{Y}) = 0 \\
&= H(\mathsf{Z}) + H(\mathsf{X}_1^n|\mathsf{Z}) && \text{chain rule} \\
&\leq nR + H(\mathsf{X}_1^n|\mathsf{Z}) && \text{by } H(\mathsf{Z}) \leq H\left(\varphi^{(n)}(\mathsf{X}_1^n)\right) \leq nR \\
&\leq nR + \hbar(P_{\mathbf{e}}) + P_{\mathbf{e}} \log|\mathcal{X}|^n && \text{by Fano's inequality for } P_{\mathbf{e}} = \mathbf{P}\left(\hat{\mathsf{X}}_1^n \neq \mathsf{X}_1^n\right) \\
&\leq nR + 1 + nP_{\mathbf{e}} \log|\mathcal{X}| && \text{by } \hbar(P_{\mathbf{e}}) \leq 1
\end{aligned}$$

Strong Converse For Fixed Length Source Coding

Lemma (Strong Converse)

For any $R < H(\mathsf{X})$ and $\epsilon > 0$, $\exists n_0$ s.t. for all $n \geq n_0$ for any rate R code s.t. $\mathbf{P}\left(\hat{\mathsf{X}}_1^n \neq \mathsf{X}_1^n\right) \geq 1 - \epsilon$.

Proof.

$$\mathcal{A}_\gamma^{(n)}(\mathsf{X}) := \{x_1^n \in \mathcal{X}_1^n : 2^{-n(H(\mathsf{X})+\gamma)} \leq p_{\mathsf{X}_1^n}(x_1^n) \leq 2^{-n(H(\mathsf{X})-\gamma)}\}$$

Recall that $\forall \delta > 0$, $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\gamma^{(n)}\right) \geq 1 - \delta$ for sufficiently large n and set $\delta = \frac{\epsilon}{2}$.

$$\mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\gamma^{(n)}(\mathsf{X})\right) \leq \frac{\epsilon}{2} \qquad \text{for sufficiently large } n \tag{1}$$

At most 2^{nR} elements of $\mathcal{A}_\gamma^{(n)}$ are decoded correctly, each with at most $2^{-n(H(\mathsf{X})-\gamma)}$ probability.

$$\begin{aligned}
\mathbf{P}\left(\{\hat{\mathsf{X}}_1^n = \mathsf{X}_1^n\} \cap \{\mathsf{X}_1^n \in \mathcal{A}_\gamma^{(n)}(\mathsf{X})\}\right) &\leq 2^{nR} 2^{-n(H(\mathsf{X})-\gamma)} \\
&= 2^{-\gamma n/2} && \text{for } \gamma = \frac{H(\mathsf{X})-R}{2} \\
&= \epsilon/2 && \text{for sufficiently large } n
\end{aligned} \tag{2}$$

$$\mathbf{P}\left(\hat{\mathsf{X}}_1^n = \mathsf{X}_1^n\right) \leq \mathbf{P}\left(\{\hat{\mathsf{X}}_1^n = \mathsf{X}_1^n\} \cap \{\mathsf{X}_1^n \in \mathcal{A}_\gamma^{(n)}(\mathsf{X})\}\right) + \mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\gamma^{(n)}(\mathsf{X})\right) \tag{3}$$

$\mathbf{P}\left(\hat{\mathsf{X}}_1^n = \mathsf{X}_1^n\right) \leq \epsilon$ for sufficiently large n by (1), (2), and (3).

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